EC4219: Software Engineering

Lecture 6 — Program Verification (1)
Formal Specifications

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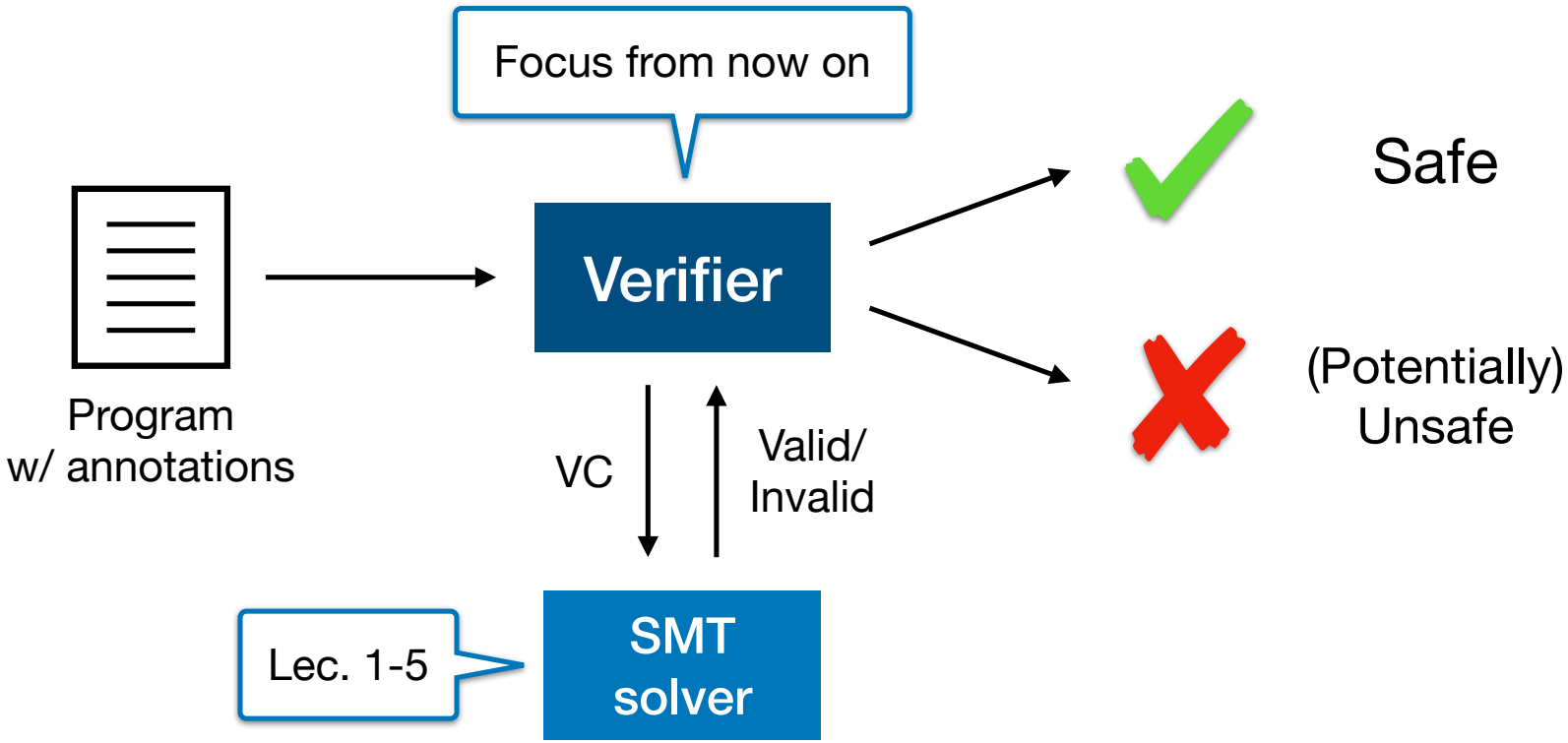
Where We Are

- In Lectures 1–5, we explored the basic internals of SMT solvers.
- In the upcoming lectures, we will treat SMT solvers as black boxes and learn basic verification techniques.



Overview: Program Verification

We will learn methods for specifying and verifying program properties.

- **Specification**: precise statement of program properties in first-order logic (FOL). We focus on specifying two forms of properties.
  - **Partial correctness properties**: If a program halts, it produces a correct output that satisfies the specification.
  - **Total correctness properties**: A program is guaranteed to terminate and produce a correct output (partial correctness + termination).
- **Verification**
  - **Inductive assertion method** for proving partial correctness
  - **Ranking function method** for proving total correctness

Specification

- A specification consists of program annotations expressed as FOL formulas.
- An annotation F at location L expresses an **invariant** asserting that F is $true$ whenever L is reached during the program execution.
- We explore three kinds of annotations.
  1. Function specification
  2. Loop invariant
  3. Assertion

Function Specification

Formulas whose free variables include the formal input and output parameters (and possibly global variables).

- **Precondition**: Specification that should be true **before entering** the function. In other words, it specifies under which inputs the function is expected to work.
- **Postcondition**: Specification that should be true **after exiting** the function. It may define the relationship between the input parameters and output parameters.

Example 1: Linear Search

```
1  @pre:
2  @post:
3  bool LinearSearch (int a[], int l, int u, int e) {
4    int i := l;
5    while (i ≤ u) {
6      if (a[i] = e) return true;
7      i := i + 1;
8    }
9    return false;
10 }
```

- @pre:
- @post: .

Example 1: Linear Search

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```

- @pre: behaves correctly only when $l \geq 0$ and $u < |a|$.
- @post: returns $true$ iff the array a contains e in the range $[l, u]$.

Example 2: Binary Search

```
1  @pre:
2  @post: rv ↔ ∃i.l ≤ i ≤ u ∧ a[i] = e
3  bool BinarySearch (int a[], int l, int u, int e) {
4    if (l > u) return false;
5    else {
6      int m := (l + u) div 2;
7      if (a[m] = e) return true;
8      else if (a[m] < e) return BinarySearch (a, m + 1, u, e)
9      else return BinarySearch (a, l, m − 1, e)
10   }
11 }
```

- @pre:
- @post: returns $true$ iff the array a contains e in the range $[l, u]$.

Example 2: Binary Search

```
1  @pre: 0 ≤ l ∧ u < |a| ∧ sorted(a, l, u)
2  @post: rv ↔ ∃i.l ≤ i ≤ u ∧ a[i] = e
3  bool BinarySearch (int a[], int l, int u, int e) {
4    if (l > u) return false;
5    else {
6      int m := (l + u) div 2;
7      if (a[m] = e) return true;
8      else if (a[m] < e) return BinarySearch (a, m + 1, u, e)
9      else return BinarySearch (a, l, m − 1, e)
10   }
11 }
```

- @pre: behaves correctly only when $l \geq 0$, $u < |a|$, and a is sorted.
- @post: returns $true$ iff the array a contains e in the range $[l, u]$.

sorted is defined in the combined theory of integers and arrays ($T_{\mathbb{Z}} \cup T_A$).

$$\textbf{sorted}(a, l, u) \iff \forall i, j.l \leq i \leq j \leq u \rightarrow a[i] \leq a[j]$$

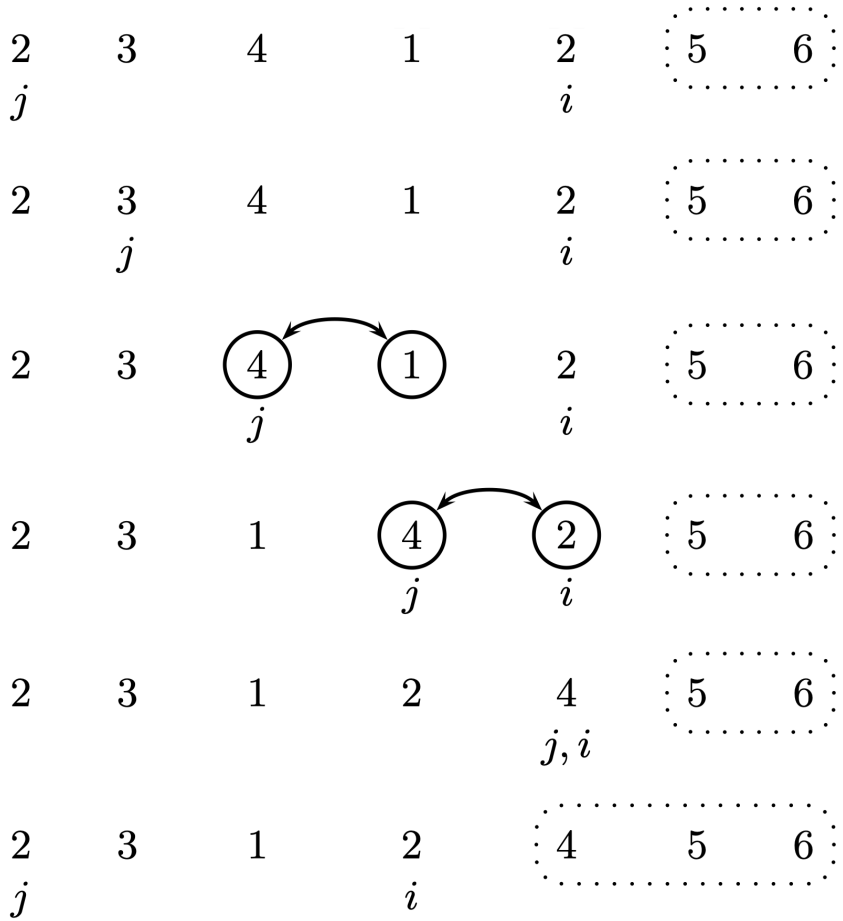
Example 3: Bubble Sort

```
1  @pre:
2  @post:
3  bool BubbleSort (int a_0[]) {
4    int[] a := a_0;
5    for (int i := |a| − 1; i > 0; i := i − 1) {
6      for (int j := 0; j < i; j := j + 1) {
7        if (a[j] > a[j + 1]) {
8          int t := a[j];
9          a[j] := a[j + 1];
10         a[j + 1] := t;
11       }
12     }
13   }
14   return a;
15 }
```

- @pre:
- @post:

Example 3: Bubble Sort

BubbleSort works by repeatedly swapping adjacent elements if they are in the wrong order, effectively "bubbling" the largest element to the end of the unsorted region.



Example 3: Bubble Sort

```
1  @pre: ⊤
2  @post: sorted(rv, 0, |rv| − 1)
3  bool BubbleSort (int a0[]) {
4    int[] a := a0;
5    for (int i := |a| − 1; i > 0; i := i − 1) {
6      for (int j := 0; j < i; j := j + 1) {
7        if (a[j] > a[j + 1]) {
8          int t := a[j];
9          a[j] := a[j + 1];
10         a[j + 1] := t;
11       }
12     }
13   }
14   return a;
15 }
```

- @pre: any array can be given as input.
- @post: the returned array is sorted.

Loop Invariant

```
1  @pre: n ≥ 0
2  @post: j = n
3  void VerySimple (int n) {
4    int i := 0;
5    int j := 0;
6    while (i ≠ n) {
7      i := i + 1;
8      j := j + 1;
9    }
10   return;
11 }
```

Q1. Does VerySimple satisfy the function specification?

Loop Invariant

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```

Q1. Does VerySimple satisfy the function specification?

Q2. Can we express lines 6–9 as a finite-length FOL formula?

Loop Invariant

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```

Q1. Does VerySimple satisfy the function specification?

Q2. Can we express lines 6–9 as a finite-length FOL formula?

We need to summarize the behaviors of loops. In our example, $i = j$ is the summarization that is precise enough to prove the partial correctness.

$$i = j \wedge i = n \rightarrow j = n$$

Loop Invariant

Each loop should be annotated with a proper loop invariant $\boldsymbol{F}$.

```
1  while
2    @F
3    (⟨condition⟩) {
4    ⟨body⟩
5  }
```

Loop invariant $\boldsymbol{F}$ is a property that (i) holds before the entrance and (ii) is preserved by executions of the loop body. That is, $\boldsymbol{F}$ holds at the beginning of every iteration. Thus,

- When entering the body, $\boldsymbol{F} \wedge \langle \boldsymbol{condition} \rangle$ holds.
- When exiting the loop, $\boldsymbol{F} \wedge \neg \langle \boldsymbol{condition} \rangle$ holds.

cf) A trivial loop invariant $\boldsymbol{true}$ holds for every loop, but it is useless in most cases.

Example 1: Linear Search

Find a nontrivial loop invariant to prove the correctness.

```
1  @pre: 0 ≤ l ∧ u < |a|
2  @post: rv ↔ ∃i.l ≤ i ≤ u ∧ a[i] = e
3  bool LinearSearch (int a[], int l, int u, int e) {
4    int i := l;
5    while
6    @L:
7    (i ≤ u) {
8      if (a[i] = e) return true;
9      i := i + 1;
10   }
11   return false;
12 }
```

Example 1: Linear Search

Find a nontrivial loop invariant to prove the correctness.

```
1  @pre: 0 ≤ l ∧ u < |a|
2  @post: rv ↔ ∃i.l ≤ i ≤ u ∧ a[i] = e
3  bool LinearSearch (int a[], int l, int u, int e) {
4    int i := l;
5    while
6    @L: l ≤ i ∧ (∀j.l ≤ j < i → a[j] ≠ e)
7    (i ≤ u) {
8      if (a[i] = e) return true;
9      i := i + 1;
10   }
11   return false;
12 }
```

- The index i is no less than l.
- We have not found an element in the previously examined indices j.

Example 2: Bubble Sort

```
1  @pre: ⊤
2  @post: sorted(rv, 0, |rv| − 1)
3  bool BubbleSort (int a[]) {
4    int[] a := a0;
5    @L1 : { −1 ≤ i < |a|
             ∧ partitioned(a, 0, i, i + 1, |a| − 1)
             ∧ sorted(a, i, |a| − 1) }
6    for (int i := |a| − 1; i > 0; i := i − 1) {
7      @L2 : { 1 ≤ i < |a| ∧ 0 ≤ j ≤ i
               ∧ partitioned(a, 0, i, i + 1, |a| − 1)
               ∧ partitioned(a, 0, j − 1, j, j)
               ∧ sorted(a, i, |a| − 1) }
8      for (int j := 0; j < i; j := j + 1) {
9        if (a[j] > a[j + 1]) {
10         int t := a[j];
11         a[j] := a[j + 1];
12         a[j + 1] := t;
13       }
14     }
15   }
16   return a;
17 }
```

$$\textbf{partitioned}(a, l_1, u_1, l_2, u_2) \iff \forall i, j. l_1 \leq i \leq u_1 < l_2 \leq j \leq u_2 \rightarrow a[i] \leq a[j]$$

Assertion

- The other formal comments on expected program behaviors.
- Usually specified by the command `assert` in most programming languages. If assertion violations occur at runtime, typically abort program executions; very useful for debugging errors.

```
1  @pre: 0 ≤ l ∧ u < |a|
2  @post: rv ↔ ∃i.l ≤ i ≤ u ∧ a[i] = e
3  bool LinearSearch (int a[], int l, int u, int e) {
4    int i := l;
5    while
6    @L : l ≤ i ∧ (∀j.l ≤ j < i → a[j] ≠ e)
7    (i ≤ u) {
8    @0 ≤ i < |a| /* expectation: array access is legal */
9      if (a[i] = e) return true;
10     i := i + 1;
11   }
12   return false;
13 }
```

Assertion

- Many kinds of assertions can be automatically generated to catch common semantic errors.
  - division-by-zero, null-dereference, accessing an array out of bounds, etc.
- For example, we can transform the C code snippet

$$\ldots;\ i := i/j;\ \ldots$$

into

$$\ldots;\ i := i/j;\ \texttt{assert}(j \neq 0);\ \ldots$$

Summary

- Goal: prove the correctness of implementations
- We explored specification methods to rigorously describe correct behavior using FOL formulas.
  1. Function specification: precondition, postcondition
  2. Loop invariant: summarization of loops
  3. Assertion: The other formal comments on expected behaviors

Next class: **Inductive assertion method** for proving partial correctness